Fachhochschule Nordwestschweiz
Hochschule für Technik

dlbs - Deep Learning Bild und Signal

Instance Segmentation of Vehicles

Technical-Report

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1 Project Statement

Accurately detecting vehicles from a driver's perspective is essential for applications such as autonomous driving, driver-assistance systems, and urban scene analysis. Instance-level predictions enable pixel-accurate delineations of each vehicle, providing a more comprehensive understanding of the scene than simple object counts.

The primary goal of this project is to develop and evaluate a system capable of accurately identifying and distinguishing between different road users, such as humans and various types of vehicles in urban street scenes.
The research questions are:

- What is the highest per-class segmentation quality achievable with a YOLO-based instance segmentation model, measured by per-class Dice and global mask mAP50–95?
- How well does a model trained primarily on German-recorded road users generalize to locally recorded Swiss road users, as measured by per-class Dice score at an IoU threshold of 0.5?
- How does the class imbalance in the training data influence per-class segmentation quality on the test set, as measured by per-class Dice and recall?
- How does instance size influence the segmentation performance of an instance segmentation model?

2 Data

We use the Cityscapes[1] *gtFine* dataset of 3,475 finely annotated street-scene images (2048×1024, forward-facing camera) from German cities. Each image carries pixel-accurate instance masks, available only for road users. We use all eight classes (*person, rider, car, truck, bus, train, motorcycle, bicycle*). We convert the Cityscapes `instanceIds` rasters to the YOLO instance-segmentation format (one normalized polygon per instance), since the raw polygons overlap occluded objects and yield noisy labels.

To study cross-regional generalization, we hold out the 122 Zürich images as a test set and the 154 Cologne images (the German city most similar to Zürich) as an indistribution reference, while the rest form train and validation.

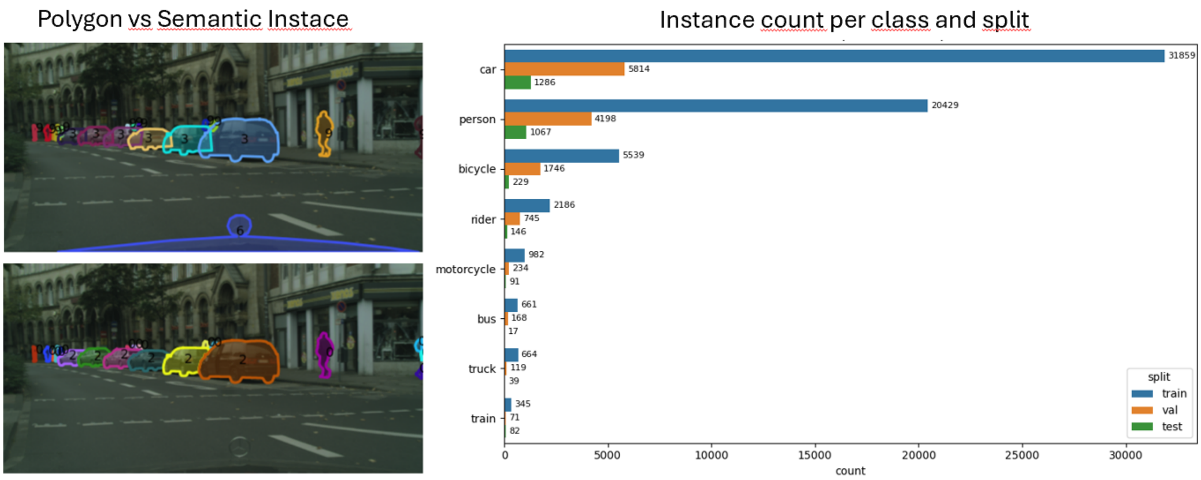

Figure 2.1: Left polygons vs pixel accurate instace masks, one can see the the overlapping masks on the two cars in the fron. Right Figure instaces per class and split

Limitations. As Figure 2.1 shows, the 78,717 instances are strongly imbalanced, with *car* and *person* alone making up about 82% and *train*, *truck* and *bus* together under 3%. Many instances are small and some masks imprecise, both hard to segment.[2]

[1] https://www.cityscapes-dataset.com/

[2] The full exploratory analysis is in the repository notebooks. Notebook 0 covers the directory layout and the YOLO label conversion, notebook 1 the image-level checks (duplicates, brightness, outliers), and notebook 2 the instance-, class- and size-level distributions, including this figure.

3 Methods

Figure 3.1 summarizes the experimental workflow. The pipeline starts with dataset preparation and label conversion, followed by training on the German Cityscapes cities and a final evaluation on a held-out test set of cities not seen during training (Zürich for the cross-region gap and Cologne as an in-distribution reference).

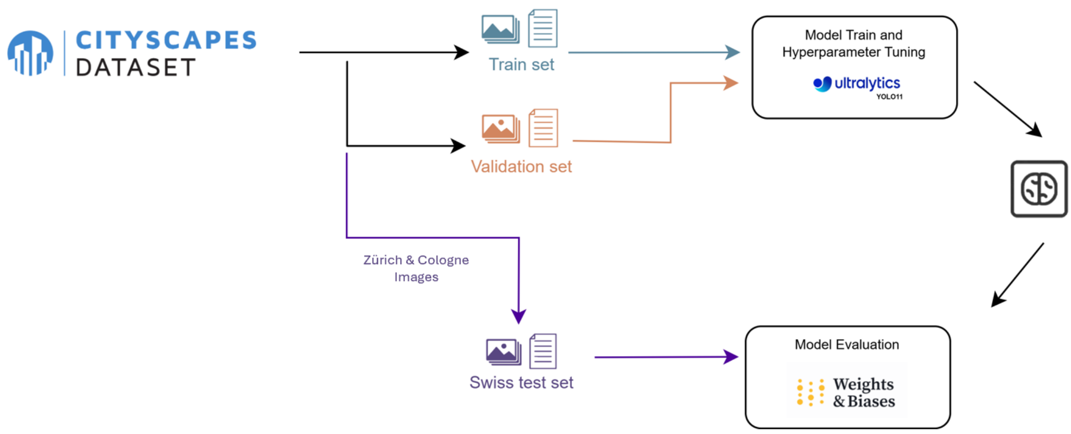

Figure 3.1: Workflow with the different subsets.

We use YOLOv11m in its segmentation configuration as the main architecture. YOLO was selected because it performs instance segmentation in a one-stage framework and is therefore suitable for real-time perception such as driver assistance, and to explore the Ultralytics YOLOv11 framework. The model is initialized with COCO-pretrained weights, which provide useful low-level and object-level features before fine-tuning.

The training strategy follows the recommendations by Karpathy and Lones Karpathy, 2019; Lones, 2024. First, we overfit a single image without augmentation to verify that the data format, labels, loss, and training loop are correct. Second, we train a baseline with fixed seeds, no augmentation, a constant learning rate, and YOLO's defaults (including weight decay 0.0005), and evaluate the final baseline across three seeds, reporting all metrics as mean $\pm$ standard error over seeds.

For evaluation we report mask mAP50–95 and Dice, both overall and per class. Accuracy is avoided because the data is imbalanced. Early stopping and checkpoint selection use only the mask fitness term, since box quality is secondary here. We additionally analyze performance stratified by city and instance size, with chi-squared tests and Holm correction to flag significant differences.

Following the recipe, after the baseline we confirm the model can overfit the full training set, then improve generalization with weight decay, street-scene augmentation, a random search over the augmentation strength, and a higher input resolution. The best configuration is selected by validation Dice.

4 Results

The full results are available as a Weights & Biases report[1] and as notebooks in the repository [2].

4.1 Most Relevant Results

We report the final test metrics across the four trained configurations (baseline, increased weight decay, street-scene augmentation, and higher input resolution). Figure 4.1 shows the global test metrics as mean $\pm$ standard error over seeds, and Figure 4.2 the per-class Dice scores. The higher-resolution configuration reached the highest values on every metric. Its global mask mAP50–95 was about 0.23, against about 0.16 for baseline and weight decay and 0.18 for augmentation. Global Dice followed the same order, with about 0.39 for baseline and weight decay, 0.43 for augmentation, and 0.50 for higher resolution. Precision and recall reached about 0.67 and 0.40 under the higher-resolution setting.

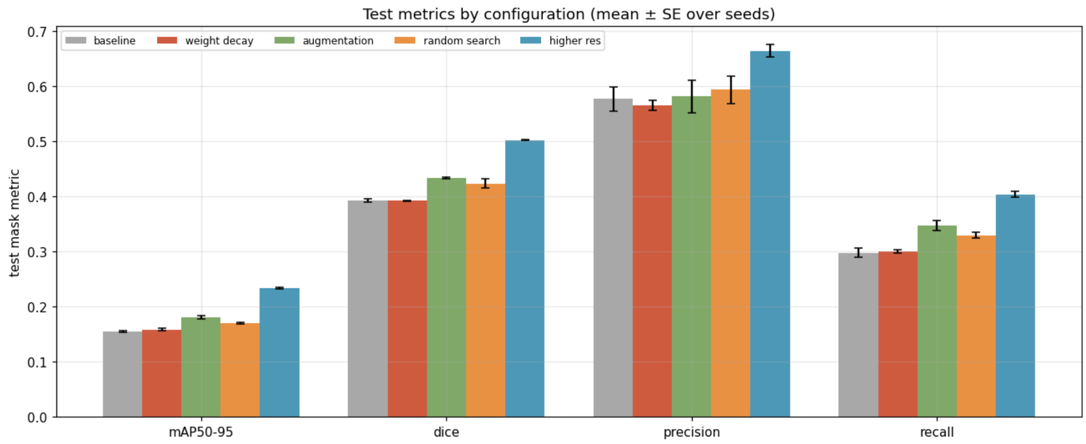

Figure 4.1: Global test metrics by configuration. Bars show mean values over seeds; error bars show the standard error.

Per-class Dice was highest under the higher-resolution configuration for every class, ranging from *truck* (0.41) to *car* (0.67), with *motorcycle* (0.55), *person* (0.53) and *rider* (0.51) above 0.50 and *bus* (0.44), *train* (0.44) and *bicycle* (0.42) below. Relative to the baseline, the largest gains were for *truck* $(0.25 \rightarrow 0.41)$, *rider* $(0.39 \rightarrow 0.51)$ and *bus* $(0.31 \rightarrow 0.44)$. Augmentation improved every per-class Dice over the baseline but stayed below the higher-resolution scores, and a random search over its copy-paste and scale parameters changed validation Dice by only about 0.01. Weight decay stayed close to the baseline on the global metrics.

Table 4.1: Highest per-class Dice scores achieved on the test set. All best values were obtained by the higher-resolution configuration.

Class	bicycle	bus	car	motorcycle	person	rider	train	truck
Dice	0.42	0.44	0.67	0.55	0.53	0.51	0.44	0.41

[1] https://wandb.ai/dlbs-jf/runs-segment/reports/Report-for-DLBS--VmlldzoxNTg1Mjg3MA
[2] https://github.com/JuanFelipeRuiz/dlbs

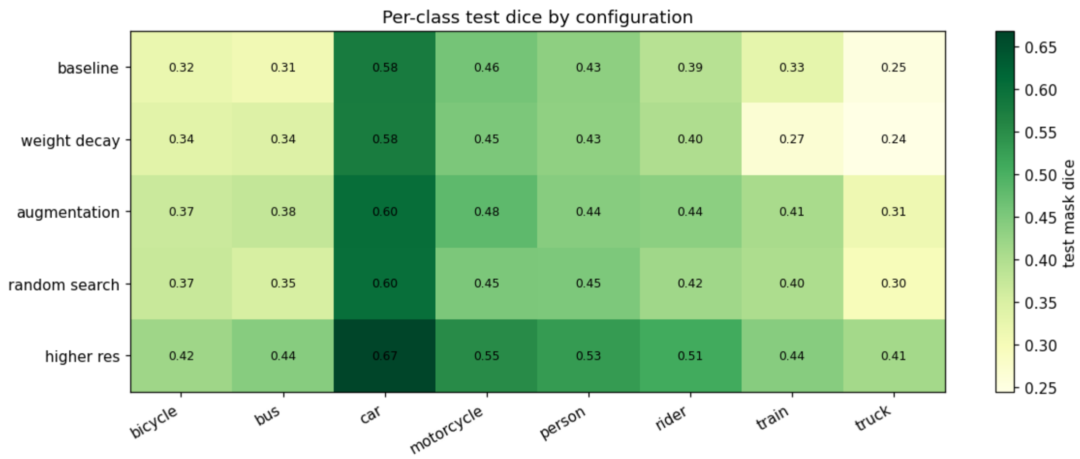

Figure 4.2: Per-class test Dice by configuration. Cell values report Dice scores for each class and model configuration.

4.2 Stratified results

We evaluate the higher-resolution model stratified by city and by instance size, using the same per-seed χ^2 test on detected versus missed instances with Cramér's V as effect size (Figure 4.3). Between cities, mask recall is 0.64 in Zürich and 0.70 in Cologne, while Dice and precision differ by less than their standard error. This city difference is significant ($p < 10^{-5}$) but small (Cramér's $V \approx 0.06$). By instance size, recall rises monotonically from 0.08 for the smallest quartile to 0.97 for the largest, a strong and significant effect ($p < 10^{-3}$, Cramér's $V \approx 0.70$). A per-class view shows that Dice does not follow class frequency closely, since the rarer *motorcycle* ($1{,}307$ instances, Dice 0.55) scores above the more frequent *bicycle* ($7{,}514$, Dice 0.42).

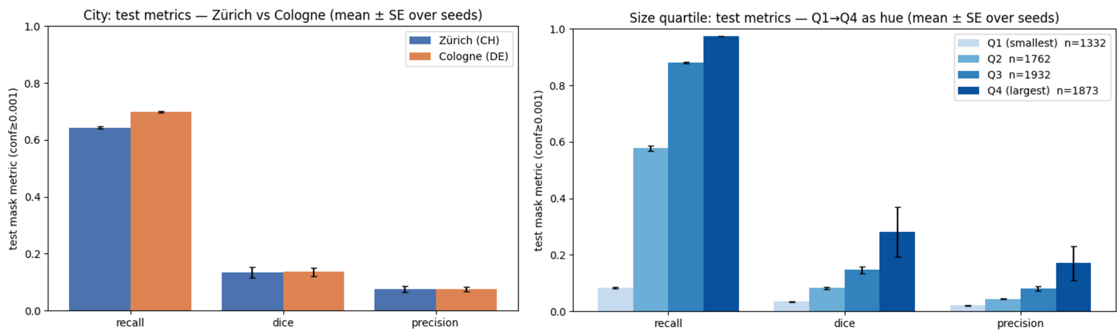

Figure 4.3: Test metrics of the higher-resolution model stratified by city (left) and by instance-size quartile (right), as mean $\pm$ standard error over seeds.

5 Discussion

We discuss the results per research question, comparing the baseline with the optimized configurations and noting where the model works and fails.

Achievable quality (RQ1). Among the four configurations, weight decay matched the baseline, augmentation gave a small gain, and only higher resolution improved performance clearly and across every class. In absolute terms the quality is still modest, and precision clearly exceeds recall (about 0.67 versus 0.40), so the model segments what it detects fairly reliably but misses a large share of instances. For the driver-perspective use case this is informative but not deployment-grade. Missing a road user directly ahead would be a safety risk, whereas a small car far in the background matters less, so the impact of the missed instances depends on the application. A follow-up step could rank or filter instances by distance and position rather than treat them equally.

Cross-regional generalization (RQ2). The city split shows almost no gap between Zürich and Cologne, so the model is not visibly biased toward the German scenes it saw more often during training. This evidence is limited, however, because Zürich is a single Swiss city recorded within the same Cityscapes setup as the training cities. It therefore probes a mild geographic shift rather than a genuinely different recording domain, for which we have no data.

Class imbalance (RQ3). The training data is strongly imbalanced, yet per-class quality does not follow class frequency closely. Recall tracks frequency, but mask quality does not, because two factors cushion the imbalance. COCO pretraining already covers most of our classes, so part of the per-class quality reflects that prior rather than our own data, and instance size is a stronger driver than frequency, with recall rising sharply from the smallest to the largest size quartile. Because about half of all instances are small, this size weakness affects a large share of the data. Class imbalance thus has a measurable but secondary effect, and the dominant failure mode is small-instance segmentation rather than rarity itself.

Limitations. The results rest on a single fixed train/test split with three seeds, which captures training variability but not data variability. Rare-class estimates (*bus* and *truck*, with a few dozen instances) carry wide uncertainty, and instances within a city or size bucket are not fully independent, so the significance tests slightly overstate certainty. RQ1 and RQ3 can be answered with confidence from the available data, whereas RQ2 holds only at the city level within Cityscapes.

6 Technical reflection

Juan Instance segmentation is quite different from image classification. The model outlines every object instead of giving one label per image. It was nice to see how YOLO does this in one stage. For a mini challenge the framework felt heavy, with many settings to tune. Next time we would build a smaller custom model without pretraining. That gives more control, though accuracy on our small dataset would likely drop.

Florian The main technical lesson is that YOLO-based instance segmentation worked for our research questions, but recall, especially for small instances, remains too low for safety-critical use. I underestimated the effort and risk of the data labelling and label conversion. Next time, I will take more time for this task itself. Furthermore, we could test more different hyperparameters in the random search for a more comprehensive study.

7 Self-Assessment

We would give ourselves a grade between 5 and 5.25. According to the evaluation rubric, we meet most criteria at a good level. The problem statement, use case, and dataset are clearly justified, the Cityscape-splits and limitations are presented transparently and the methodology follows a clear approach to baseline, overfitting, regularization, and tuning. The evaluation is featuring Dice, mAP, multiple seeds, class-, city-, and size-based analyses, as well as a critical discussion. We see some drawbacks in the limited scope of hyperparameter tuning, the only mild domain shift between Zurich and Cologne, and the recall performance. Overall, the work is solid, but not entirely excellent.

Literatur

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